Derivation of slant height 'l'

ABC is a right angled triangle, right angled at B. The hypotenuse, base and height of the triangle are represented by l, r and h respectively.

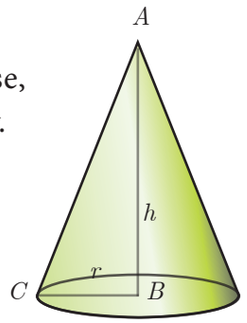


Fig. 7.12

Now, using Pythagoras theorem in ΔABC,

$$AC^2 = AB^2 + BC^2$$
$$l^2 = h^2 + r^2$$
$$l = \sqrt{h^2 + r^2} \text{ units}$$

(ii) Total surface area

Total surface area of a cone = C.S.A. + base area of the cone

$= \pi rl + \pi r^2$ ($\because$ the base is a circle)

T.S.A. of a right circular cone $= \pi r(l + r)$ sq. units.

Example 7.5 The radius of a conical tent is 7 m and the height is 24 m. Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is 4 m?

Solution Let r and h be the radius and height of the cone respectively.

Given that, radius $r = 7$ m and height $h = 24$ m

Hence, $l = \sqrt{r^2 + h^2}$

$= \sqrt{49 + 576}$

$l = \sqrt{625} = 25$ m

C.S.A. of the conical tent $= \pi rl$ sq. units

Area of the canvas $= \frac{22}{7} \times 7 \times 25 = 550$ m^2

Now, length of the canvas $= \frac{\text{Area of the canvas}}{\text{width}} = \frac{550}{4} = 137.5$ m

Therefore, the length of the canvas is137.5 m

Example 7.6 If the total surface area of a cone of radius 7cm is 704 cm^2, then find its slant height.

Solution Given that, radius $r = 7$ cm

Now, total surface area of the cone $= \pi r\left(l + r\right)$ sq. units

T.S.A. $= 704$ cm^2

$704 = \frac{22}{7} \times 7\left(l + 7\right)$

$32 = l + 7 \Rightarrow l = 25$ cm

Therefore, slant height of the cone is 25 cm.

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